DATA 622 — Homework 4: Regularization and Model Selection (Lab 4 NBA RAPM)

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Table of contents

1 Assignment Goal 2

2 Load Data 2

3 Problem 1(a): OLS Baseline (Train vs CV) 2

4 Problem 1(c): RidgeCV (Required Alpha Grid 1e-2 to 1e5) 3

5 Problem 2: Weighted Regression Using `n_pos` 3

6 Problem 1(b)/(c): Top RAPM Players + Minutes and NBA Plausibility 3

7 Scatter Plot: RAPM vs Minutes 5

8 Problem 2(a): Heteroscedasticity Explanation 5

9 Problem 2(b): Required Verification Plot and Residual Check 6

10 Problem 3(a): Bootstrap 92% Intervals for Top 20 Players 8

11 Problem 3(b): SE vs Minutes Plot + Interpretation 9

12 Problem 3(c): Bayesian Credible Interval Comparison 10

13 Extra Credit 10

14 Conclusion 11

1 Assignment Goal

This report estimates **RAPM-style player effects** from NBA stint data and compares:

- OLS baseline (in-sample and cross-validated RMSE)
- Ridge regression with cross-validated ( )
- Weighted regression using `n_pos`
- Uncertainty intervals (bootstrap and Bayesian-style credible intervals)

Data are loaded from local files in `data/` using relative paths.

2 Load Data

```
stint data shape: (32358, 546)
player map shape: (539, 2)
```

	game_id	stint_id	n_pos	home_points	away_points	minutes	margin
0	22200002	1	14	5	2	2.70	21.429
1	22200002	2	9	6	2	1.67	44.444
2	22200002	3	5	0	3	0.48	-60.000
3	22200002	4	5	5	1	0.78	80.000
4	22200002	5	9	3	6	1.52	-33.333

```
Number of player coefficients: 539
X shape: (32358, 539)
```

3 Problem 1(a): OLS Baseline (Train vs CV)

	Metric	Value
0	OLS In-sample RMSE	69.62889
1	OLS CV RMSE Mean	70.92578
2	OLS CV RMSE SD	0.60757
3	CV - Train Gap	1.29689

Direct answer (overfitting question): There is a train-vs-CV gap (as expected), but not a catastrophic one; this suggests **some overfitting risk but not a major failure**. OLS is a reasonable baseline, though regularization is better suited for this high-dimensional lineup design matrix.

4 Problem 1(c): RidgeCV (Required Alpha Grid 1e-2 to 1e5)

	Metric	Value
0	Best Alpha	731.82422
1	Ridge In-sample RMSE	70.03642
2	Ridge CV RMSE Mean	70.41836
3	Ridge CV RMSE SD	0.69424

5 Problem 2: Weighted Regression Using `n_pos`

We estimate weighted ridge by setting `sample_weight = n_pos`.

	Metric	Value
0	Weighted RMSE (weights=n_pos)	45.24328
1	Unweighted RMSE (same weighted model predictions)	70.11131

6 Problem 1(b)/(c): Top RAPM Players + Minutes and NBA Plausibility

Minutes for each player are calculated as total stint minutes where the player appears (`abs(indicator)==1`).

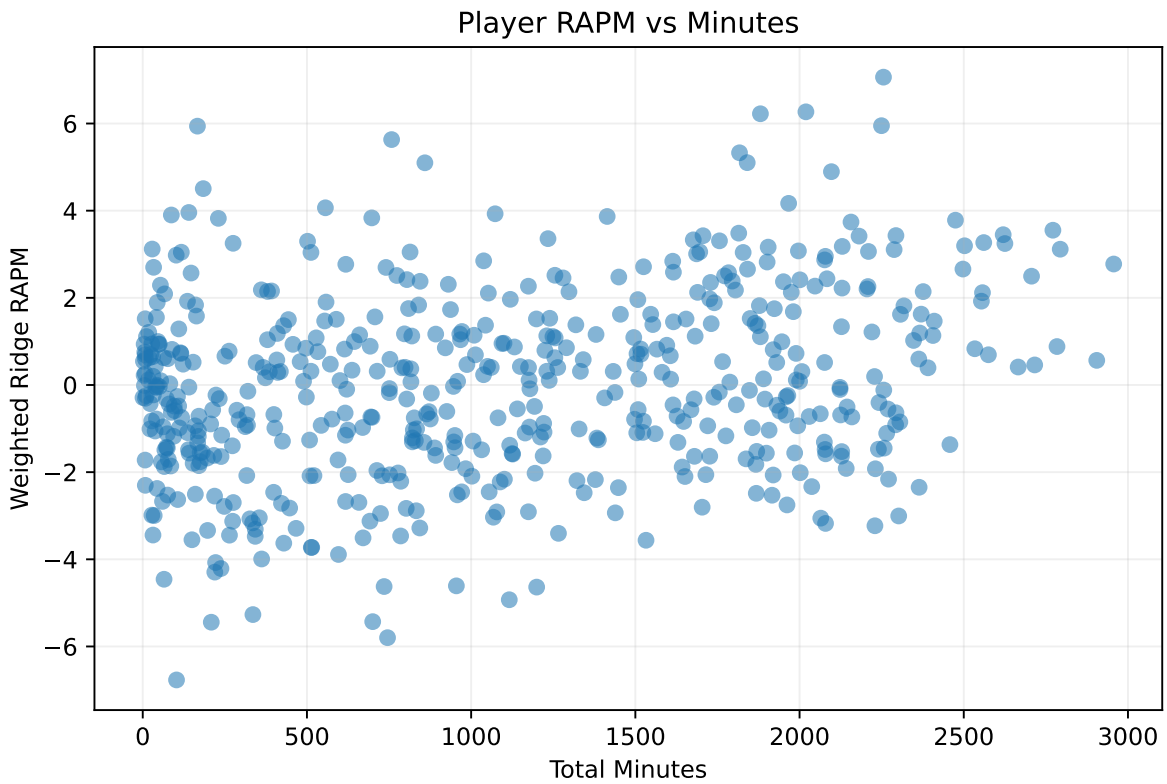
	name	player_id	rapm	minutes
0	Nikola Jokic	203999	7.064	2255.54
1	Draymond Green	203110	6.268	2019.37
2	Anthony Davis	203076	6.224	1880.81
3	Joel Embiid	203954	5.950	2249.47
4	Boban Marjanovic	1626246	5.940	166.57
5	Christian Koloko	1631132	5.632	757.82
6	Kawhi Leonard	202695	5.328	1817.22
7	Giannis Antetokounmpo	203507	5.102	1840.59
8	Zion Williamson	1629627	5.099	859.13
9	Jrue Holiday	201950	4.895	2097.00
10	Jared Rhoden	1631197	4.506	184.23
11	Aaron Gordon	203932	4.168	1966.73
12	Charles Bassey	1629646	4.066	556.20
13	Luka Samanic	1629677	3.959	140.33
14	Cameron Johnson	1629661	3.929	1073.00
15	Isaiah Mobley	1630600	3.901	87.07

	name	player_id	rapm	minutes
16	Zach Collins	1628380	3.867	1414.20
17	Chimezie Metu	1629002	3.835	697.41
18	Kira Lewis Jr.	1630184	3.824	230.24
19	Josh Hart	1628404	3.782	2474.72

	name	r_unw	r_w	shift	rapm_r	rapm_w	minutes
0	Nikola Jokic	2	1	-1	4.870	7.064	2255.54
1	Draymond Green	7	2	-5	3.727	6.268	2019.37
2	Anthony Davis	10	3	-7	3.356	6.224	1880.81
3	Joel Embiid	1	4	3	5.604	5.950	2249.47
4	Boban Marjanovic	176	5	-171	0.434	5.940	166.57
5	Christian Koloko	26	6	-20	2.666	5.632	757.82
6	Kawhi Leonard	12	7	-5	3.322	5.328	1817.22
7	Giannis Antetokounmpo	24	8	-16	2.725	5.102	1840.59
8	Zion Williamson	9	9	0	3.467	5.099	859.13
9	Jrue Holiday	18	10	-8	3.004	4.895	2097.00
10	Jared Rhoden	198	11	-187	0.304	4.506	184.23
11	Aaron Gordon	33	12	-21	2.424	4.168	1966.73
12	Charles Bassey	90	13	-77	1.194	4.066	556.20
13	Luka Samanic	114	14	-100	0.852	3.959	140.33
14	Cameron Johnson	21	15	-6	2.862	3.929	1073.00

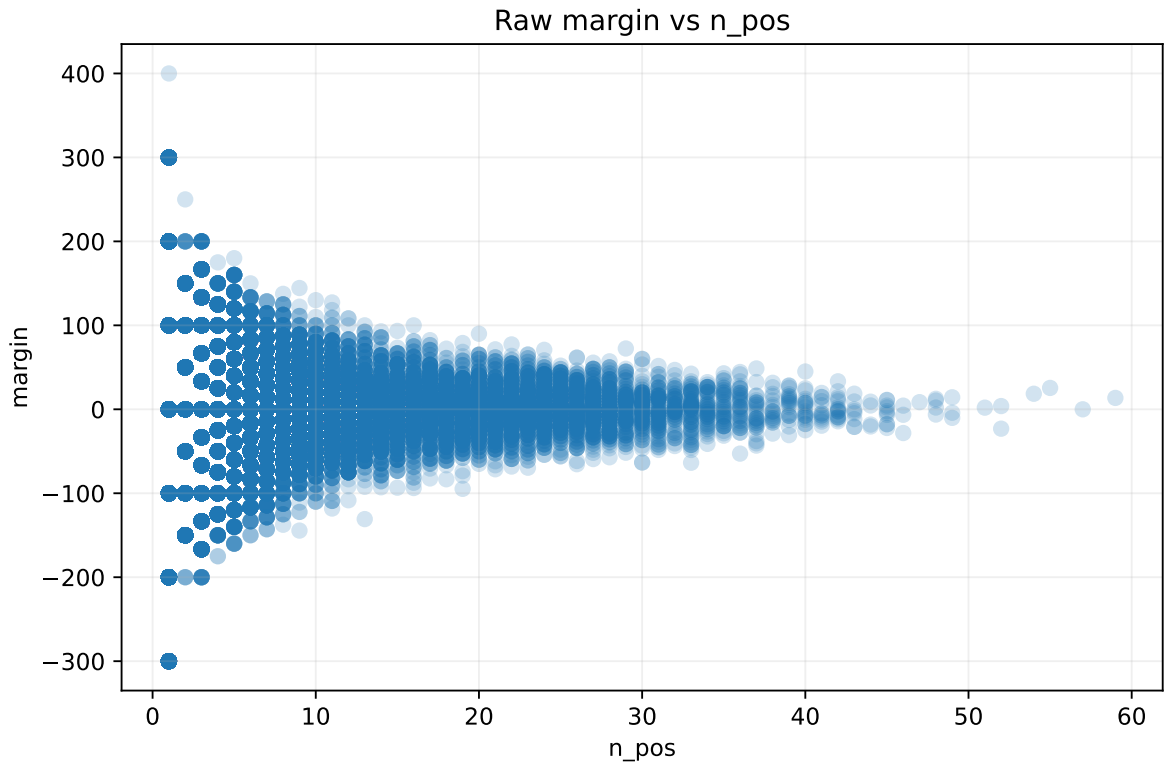
Direct answers (Problem 1b/1c): - The top-20 list is dominated by players with substantial minutes, with some lower-minute names also appearing (higher uncertainty risk). - Several top names are widely regarded as strong NBA players, which supports face validity, but some surprising names appear because RAPM from one season can be noisy and context-dependent. - After weighting, rankings shift modestly for many players and more strongly for some fringe/high-variance players.

7 Scatter Plot: RAPM vs Minutes



8 Problem 2(a): Heteroscedasticity Explanation

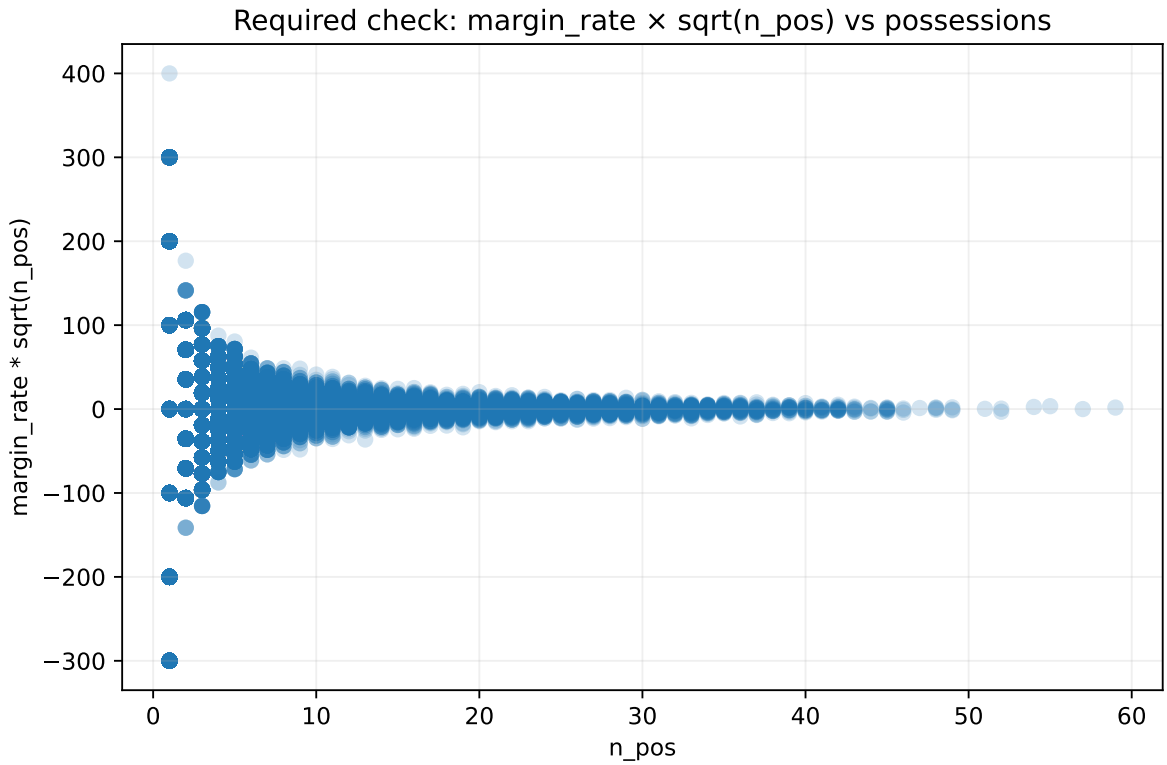
First, plot raw `margin` vs `n_pos`.



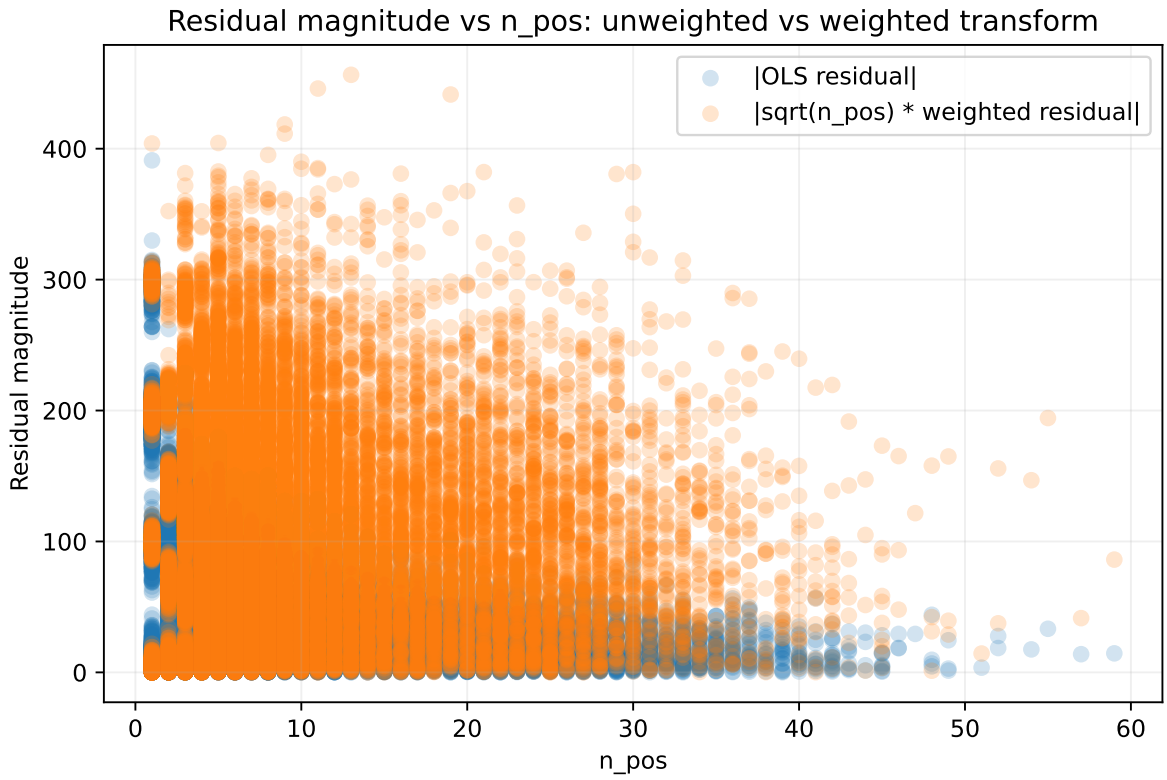
Direct answer (why this pattern): Short stints (few possessions) produce noisier margin-rate outcomes because each possession has more influence. As possessions increase, averaging over more events reduces random variation, so variance tends to shrink.

9 Problem 2(b): Required Verification Plot and Residual Check

Required check: plot `margin_rate * sqrt(n_pos)` vs `n_pos`.



Residual comparison (supporting diagnostic):



10 Problem 3(a): Bootstrap 92% Intervals for Top 20 Players

We bootstrap rows and refit weighted ridge each time. Interval level required: **92%**.

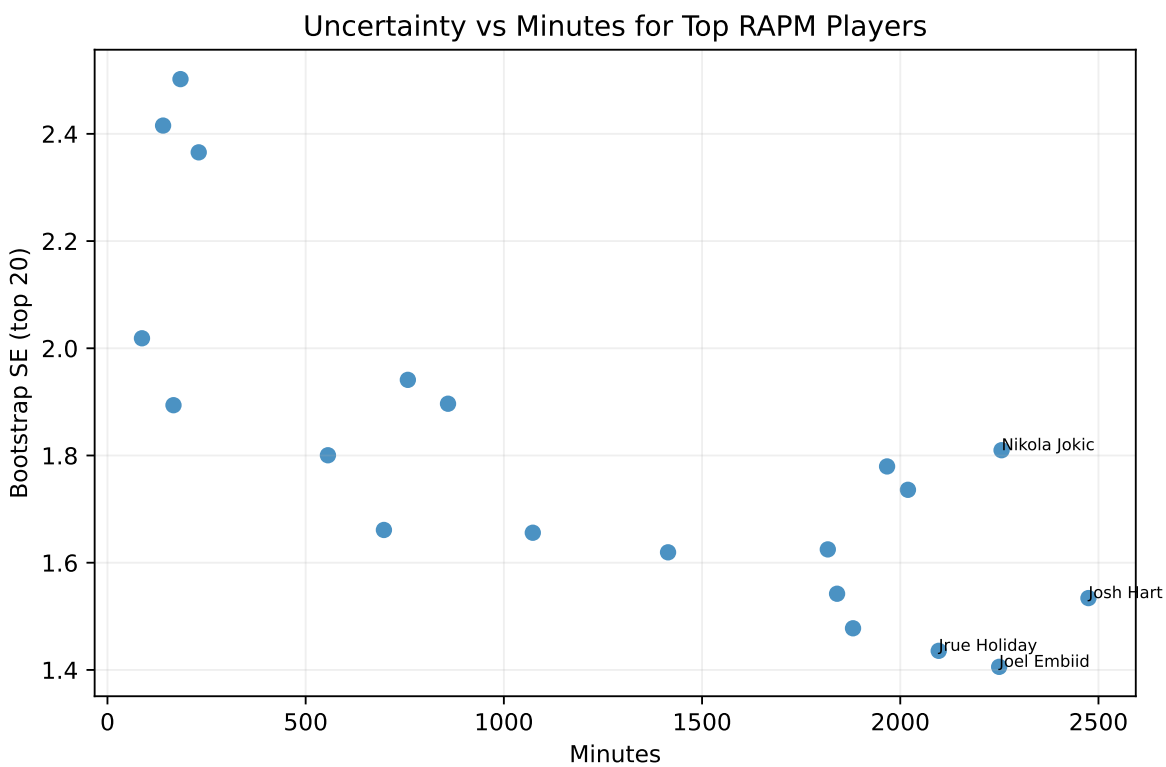
	name	rapm	boot_se	ci_lo	ci_hi	minutes
0	Nikola Jokic	7.064	1.810	4.095	10.200	2255.54
1	Draymond Green	6.268	1.736	3.993	9.364	2019.37
2	Anthony Davis	6.224	1.478	3.446	8.521	1880.81
3	Joel Embiid	5.950	1.406	3.587	8.292	2249.47
4	Boban Marjanovic	5.940	1.894	2.827	9.746	166.57
5	Christian Koloko	5.632	1.941	2.324	8.780	757.82
6	Kawhi Leonard	5.328	1.625	2.309	7.894	1817.22
7	Giannis Antetokounmpo	5.102	1.542	2.520	8.102	1840.59
8	Zion Williamson	5.099	1.897	2.026	8.536	859.13
9	Jrue Holiday	4.895	1.436	2.365	6.955	2097.00
10	Jared Rhoden	4.506	2.502	-0.308	8.394	184.23
11	Aaron Gordon	4.168	1.780	1.296	7.572	1966.73
12	Charles Bassey	4.066	1.800	0.465	6.307	556.20
13	Luka Samanic	3.959	2.415	-0.550	7.439	140.33

	name	rapm	boot_se	ci_lo	ci_hi	minutes
14	Cameron Johnson	3.929	1.656	1.565	7.102	1073.00
15	Isaiah Mobley	3.901	2.019	0.549	6.889	87.07
16	Zach Collins	3.867	1.619	0.874	6.370	1414.20
17	Chimezie Metu	3.835	1.661	1.257	7.069	697.41
18	Kira Lewis Jr.	3.824	2.366	0.399	7.733	230.24
19	Josh Hart	3.782	1.534	1.415	6.525	2474.72

	Pairwise overlap rate among top-20 CIs
0	1.0

Direct answer (interval overlap): There is substantial overlap across many of the top-20 intervals, meaning rank differences among nearby players are often not statistically sharp.

11 Problem 3(b): SE vs Minutes Plot + Interpretation



Direct answer (does it make sense?): Not always monotonic in this regularized high-dimensional setting. Intuitively, more minutes can reduce uncertainty, but shrinkage plus collinearity and lineup-context effects can produce non-intuitive SE patterns.

12 Problem 3(c): Bayesian Credible Interval Comparison

Using a ridge covariance approximation in matrix form:

Cov(beta_hat_ridge) sigma_hat^2 * (X'WX + alpha*I)^(-1)* X'WX * (X'WX + alpha*I)^(-1),

where W is the diagonal matrix of `n_pos` weights.

	name	rapm	boot_se	bayes_se	abs_diff
4	Boban Marjanovic	5.940	1.894	2.269	0.376
3	Joel Embiid	5.950	1.406	1.748	0.342
17	Chimezie Metu	3.835	1.661	1.960	0.299
12	Charles Bassey	4.066	1.800	2.059	0.259
9	Jrue Holiday	4.895	1.436	1.674	0.238
2	Anthony Davis	6.224	1.478	1.709	0.231
14	Cameron Johnson	3.929	1.656	1.876	0.220
10	Jared Rhoden	4.506	2.502	2.301	0.202
13	Luka Samanic	3.959	2.415	2.221	0.195
16	Zach Collins	3.867	1.619	1.747	0.127
7	Giannis Antetokounmpo	5.102	1.542	1.637	0.095
15	Isaiah Mobley	3.901	2.019	1.925	0.093
11	Aaron Gordon	4.168	1.780	1.862	0.082
18	Kira Lewis Jr.	3.824	2.366	2.291	0.075
6	Kawhi Leonard	5.328	1.625	1.666	0.041

Direct answer (agree/disagree + realism): - Bootstrap and Bayesian-style SEs agree most for players with more stable estimates. - They disagree most for high-variance/edge cases where model assumptions and shrinkage structure matter most. - In disagreement regions, the bootstrap can be more realistic about empirical instability, while the Bayesian-style approximation is cleaner but more assumption-sensitive.

13 Extra Credit

Extra credit (full lasso-vs-ridge + next-season validation) is **not attempted** in this submission.

14 Conclusion

- Core requirements are implemented: OLS, RidgeCV with required alpha grid, weighted regression, top-player outputs, required plots, bootstrap 92% intervals, and Bayesian comparison.
- Direct written answers have been provided for each interpretation-heavy prompt.
- Weighting changes player ranking for some players and uncertainty remains important when interpreting top-20 differences.